

Section 2E. Model Quality

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Assessing Model Quality

We should differentiate between two types of data:

- ▶ **Training dataset** $\mathcal{D}_{\text{Tr}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$: Data available while estimating the unknown parameters θ of your parametric model, $\hat{f}(\mathbf{x}; \theta)$. The **training MSE** is defined as

$$\text{MSE}_{\text{Tr}} = \frac{1}{N} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{Tr}}} \left(y_i - \hat{f}(\mathbf{x}_i; \theta) \right)^2$$

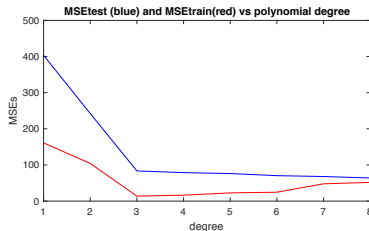
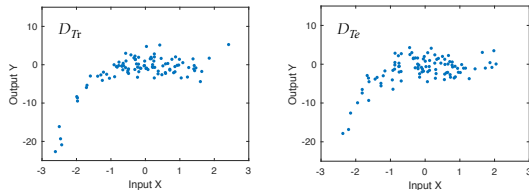
- ▶ **Testing dataset** $\mathcal{D}_{\text{Te}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^M$: Data that your learning algorithm has *not* seen during training and can be used to estimate the performance of future predictions using the **test MSE**

$$\text{MSE}_{\text{Te}} = \frac{1}{M} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{Te}}} \left(y_i - \hat{f}(\mathbf{x}_i; \theta) \right)^2$$

The test MSE is a better reflection of the performance of your model.

Model Selection

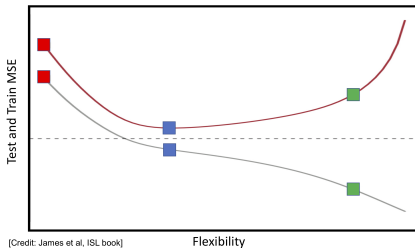
Consider a training and a testing datasets, $\mathcal{D}_{Tr}$ and $\mathcal{D}_{Te}$ containing samples $(\mathbf{x}_i, y_i) \sim f_{X,Y}$

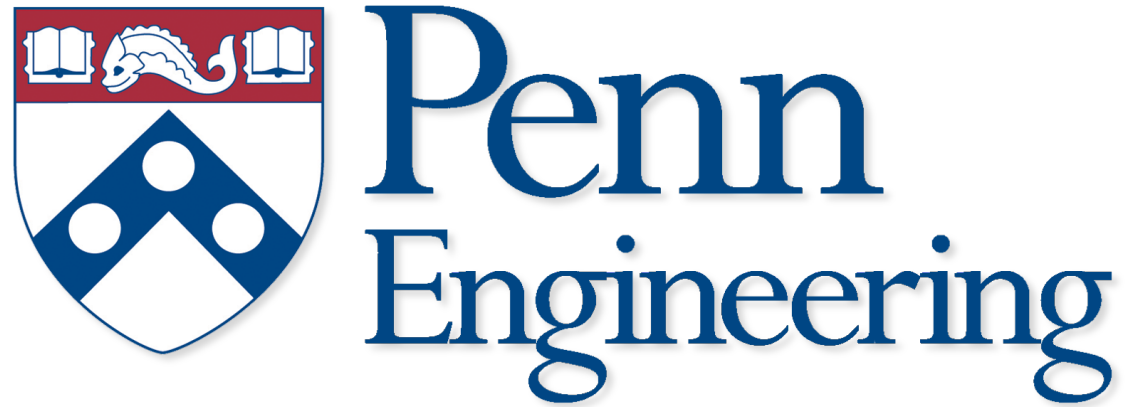


Model Selection (cont.)

Comments on test and train MSE (conceptual figure below):

- ▶ **Train MSE** (gray plot) decays monotonically as the flexibility increases.
- ▶ **Test MSE** (red plot) presents an optimal minimum value (blue square).
 - ▶ Below the optimal flexibility level, the model $\hat{f}$ is not flexible enough to learn f faithfully (red square)
 - ▶ Above the optimal flexibility level, the model $\hat{f}$ is too flexible and starts following the noise in our training data (green square)





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